

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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SCIENCE

0893/01

Paper 1

October 2023

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

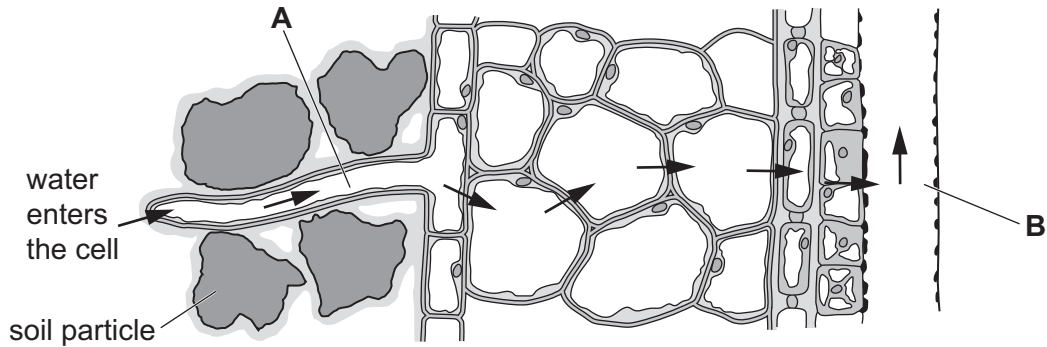
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

- 1 The arrows show the pathway of water through the root.



- (a) Name the type of root cell labelled **A**.

..... [1]

- (b) Suggest **one** way root cell **A** is adapted to its function.

Use the diagram to help you.

..... [1]

- (c) Water is transported from the roots to the stem and the leaves through vessel **B**.

Name vessel **B**.

..... [1]

- (d) Vessel **B** transports minerals to the leaves.

Name the mineral needed to make chlorophyll.

..... [1]

2 Safia has a solution of copper sulfate.

Describe how Safia makes **large** and **dry** copper sulfate crystals from this solution.

Use labelled diagrams to help you with your description.

.....

.....

.....

.....

.....

.....

.....

[3]

3 Heat and temperature are different.**(a)** Tick (✓) **all** the correct statements that describe heat.

a measure of how cold something is

☐

a measure of how hot something is

☐

a measure of the thermal energy of a substance

☐

measured in MJ

☐

transferred by conduction

☐

[2]

(b) Measurements of temperature in °C can be positive, negative or zero.

Describe how the temperature measurements of water can be positive, negative or zero.

Use ideas about the three states of matter in your answer.

.....

.....

.....

.....

.....

[2]

4 This question is about the carbon cycle and climate change.

(a) Complete the carbon cycle diagram.

Choose words from the list.

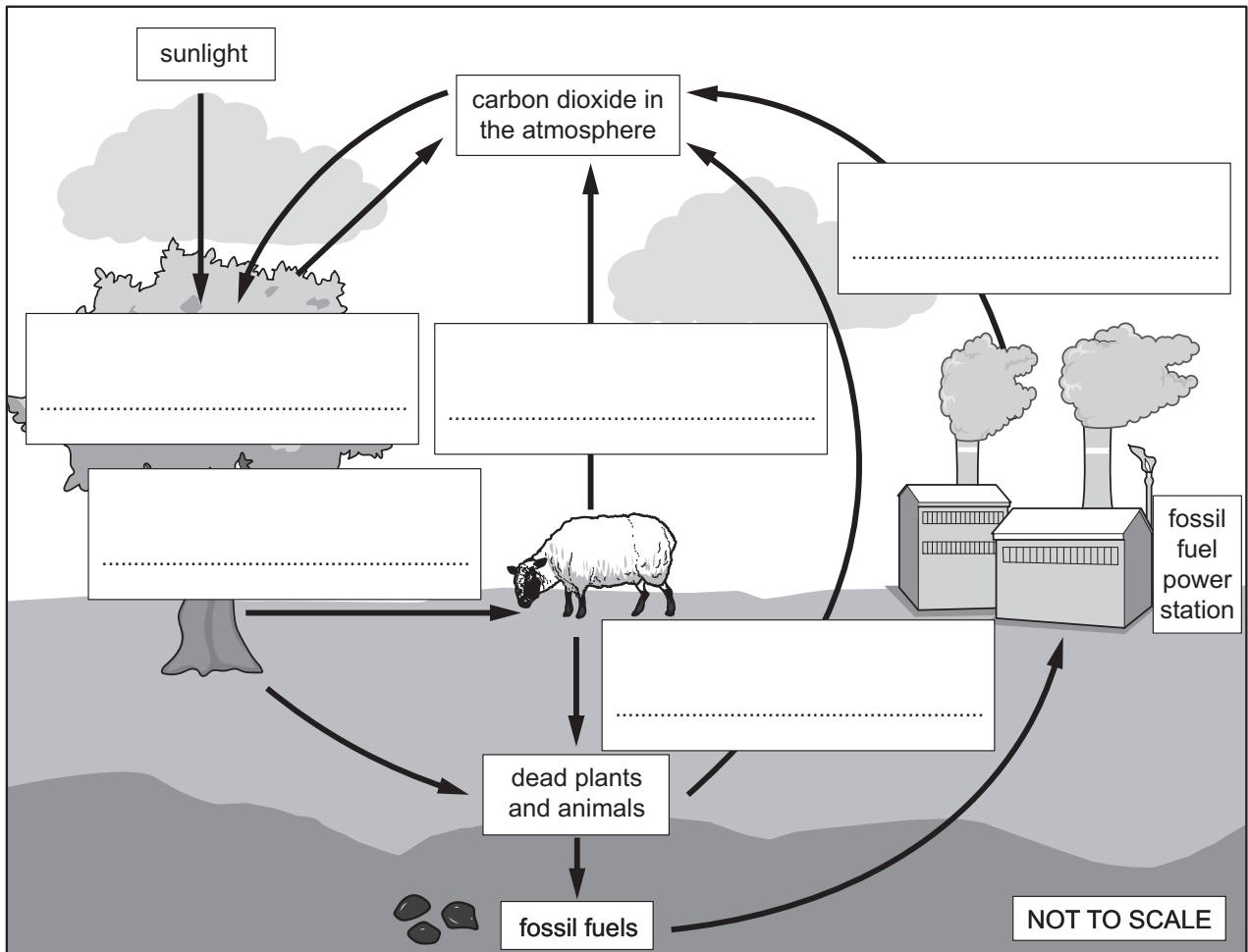
combustion

decomposition

feeding

photosynthesis

respiration



[3]

(b) Describe what happens during decomposition.

.....
 [1]

(c) The main cause of climate change is burning fossil fuels.

Describe **one** effect of climate change.

.....
 [1]

- 5 Mia and Aiko are discussing ideas about fetal development and the health of the mother.



Mia

A scientific study has shown that the mass of an unborn baby will be less if the mother has an unhealthy diet.

So, I think the mother should be careful what she eats.



Aiko

My mother says she ate what she wanted and I was born healthy.

So, I don't think it matters what the mother eats.

- (a) Who has made a more informed decision?

Tick (✓) **one** box.

Mia

☐

Aiko

☐

Explain your answer.

.....

.....

[1]

- (b) The diet of the mother is one factor that affects fetal development.

Write down **one other** factor that affects fetal development.

.....

[1]

6 Many scientists believe that a large asteroid collided with the Earth over 65 million years ago.

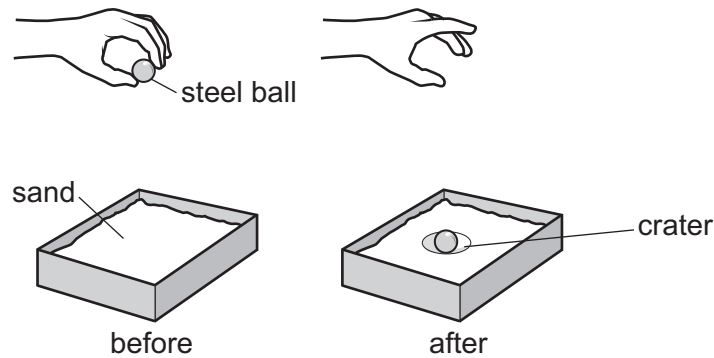
(a) Two effects of this collision were the formation of a crater and climate change.

Write down **one other** possible effect of this collision.

..... [1]

(b) Carlos models an asteroid colliding with the Earth.

He drops a steel ball into a box containing sand.



(i) Describe what the steel ball and the sand represent in his model.

steel ball

sand

[2]

(ii) Suggest **two** ways Carlos increases the width and depth of the crater.

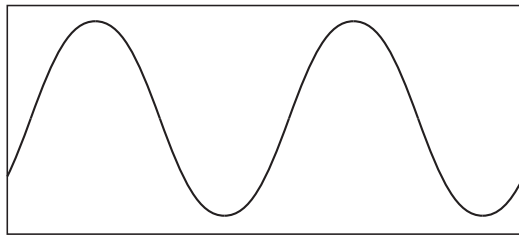
1

2

[2]

7 Waveforms show how sound waves interact.

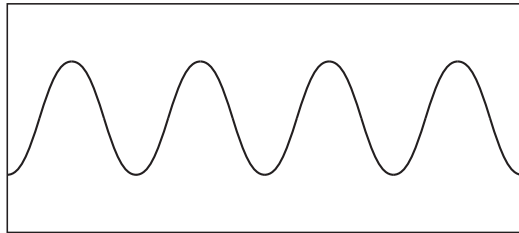
Look at the different waveforms.



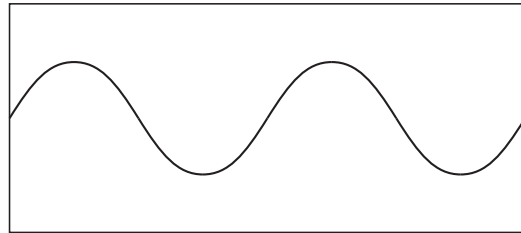
A



B



C



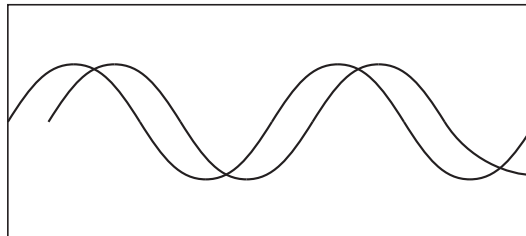
D

- (a)** Waveform **B** is the result of the interaction between two waveforms.

Which word describes this interaction?

..... [1]

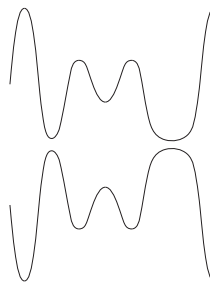
- (b)** Look at this diagram showing two waveforms.



Which waveform **A**, **B**, **C** or **D** is the result of the interaction between these two waveforms?

..... [1]

- (c)** Look at this diagram showing two waveforms.



Which waveform **A**, **B**, **C** or **D** is the result of the interaction between these two waveforms?

..... [1]

8 (a) Chen investigates the inheritance of sex in humans.

Chen:

- uses green balls to represent gametes with X chromosomes
- uses yellow balls to represent gametes with Y chromosomes
- puts 25 green balls and 25 yellow balls into bag **A**
- puts 50 green balls into bag **B**
- takes one ball out of each bag and records the colours
- returns the balls to their bags.

Explain why bag **A** represents male gametes.

.....
 [1]

(b) Chen takes balls out of the bags five more times.

(i) Look at the table of his results.

The table is **not** complete.

Complete the table to identify if the two balls (gametes) chosen represent a male or a female offspring.

	colour of ball from bag A	colour of ball from bag B	male or female offspring
1	green	green	
2	yellow	green	
3	green	green	
4	yellow	green	
5	green	green	
6	green	green	

[1]

(ii) The chance of a male and a female producing a male offspring is 50%.

Chen's results do **not** show a value of 50%.

Suggest why.

.....
 [1]

- 9 Sodium is in Group 1 of the Periodic Table.

Sodium reacts with chlorine to form an ionic compound.

This ionic compound contains sodium ions, Na^+ , and chloride ions, Cl^- .

Why is a sodium ion positively charged?

Circle the correct answer.

a sodium atom gains a proton

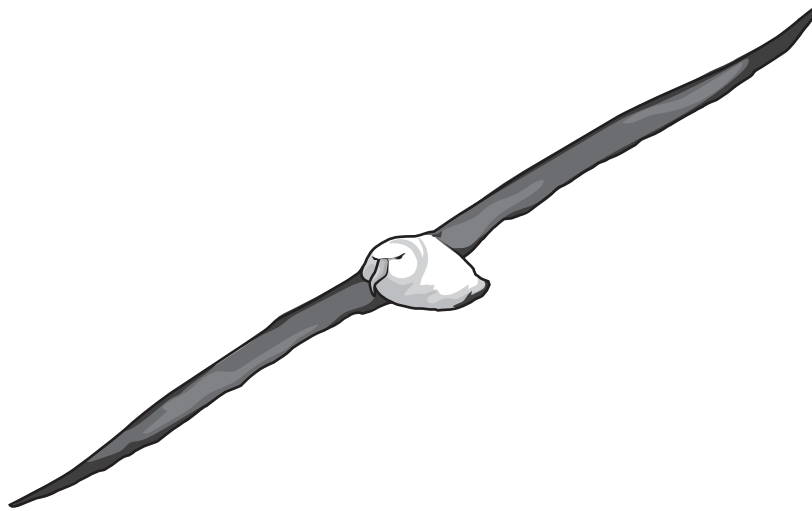
a sodium atom loses a proton

a sodium atom gains an electron

a sodium atom loses an electron

[1]

- 10 The albatross is a sea bird with very long wings.



- (a) The albatross may have long wings due to the process of natural selection.

Tick (✓) **all** the boxes that show correct statements about natural selection and the albatross.

The albatross with longer wings pass on their genes to their offspring.

☐

There is **no** variation in the length of albatross wings.

☐

The albatross do **not** compete for food.

☐

The length of the wings decreased over many generations.

☐

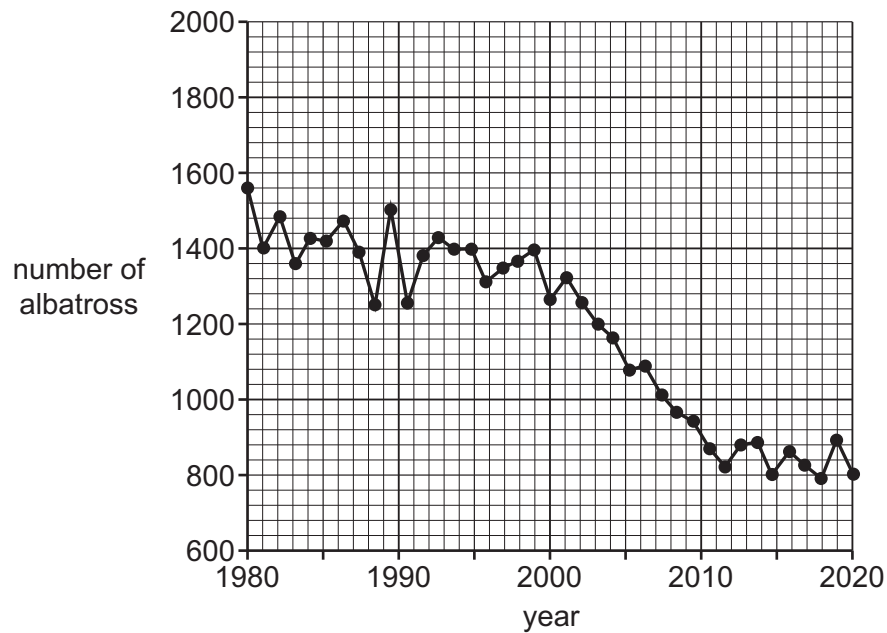
The albatross with longer wings are more likely to survive.

☐

[2]

(b) A scientist counts the number of albatross that nest on an island every year.

Look at the results.



Complete these sentences.

Between the years 1980 and 2020, the number of albatross

One reason for this could be

If the trend in the graph continues, the albatross species could become

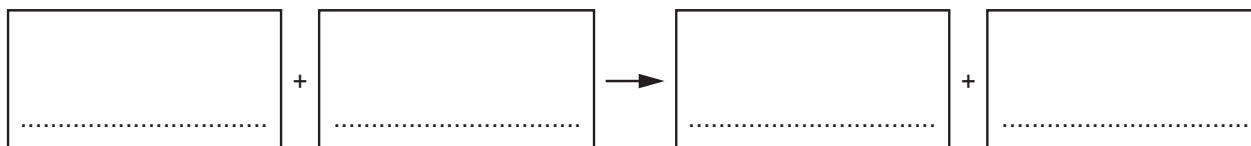
.....

[3]

11 Rajiv investigates the reaction of zinc, Zn, and dilute sulfuric acid, H_2SO_4 .

Hydrogen, H_2 , and zinc sulfate, ZnSO_4 , are made in the reaction.

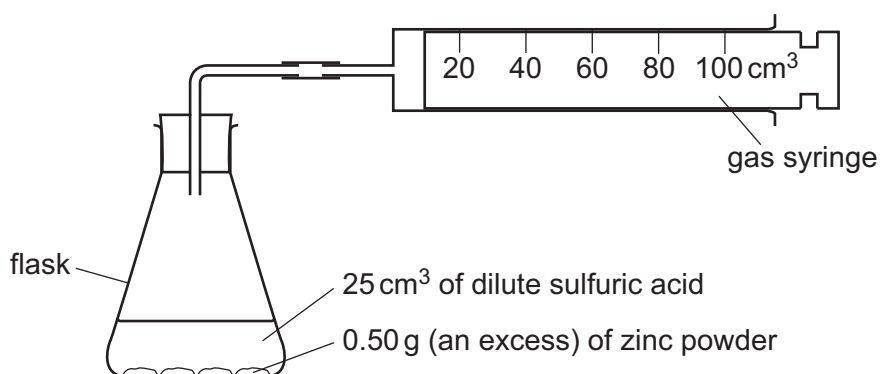
(a) Write the symbol equation for this reaction.



[2]

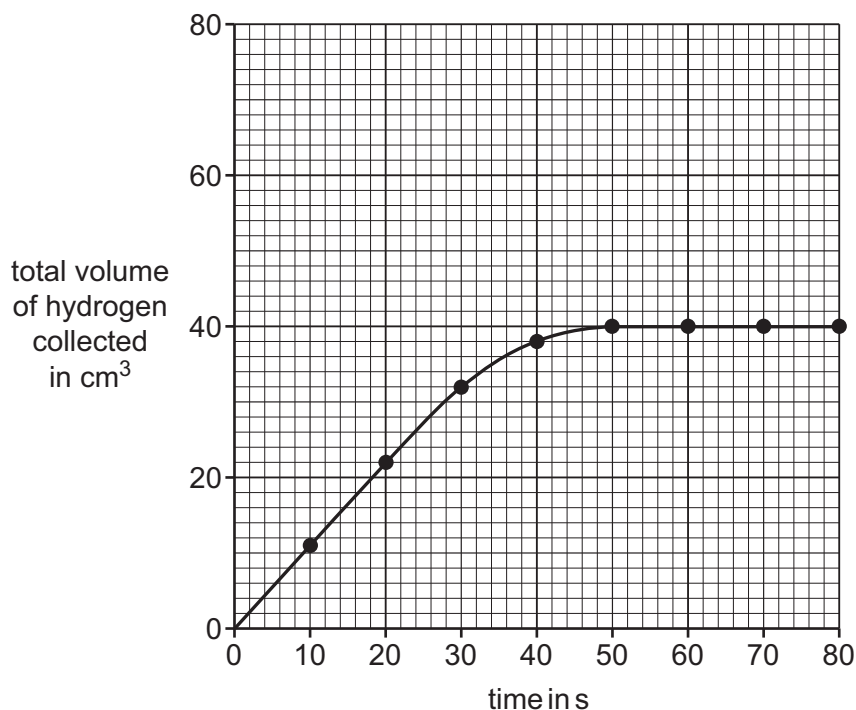
(b) Rajiv:

- puts 25 cm^3 of dilute sulfuric acid into a flask
- adds 0.50 g (an excess) of zinc powder to the sulfuric acid
- assembles the equipment shown in the diagram



- measures the total volume of hydrogen collected in the gas syringe every 10 seconds.

(i) Look at the graph showing his results.



Rajiv repeats the experiment using the same quantities of zinc powder and dilute sulfuric acid.

He uses acid at a **higher** temperature.

He does **not** change any other conditions.

Sketch a line on the graph to show the results of this experiment.

Label the line **H**.

[2]

(ii) Rajiv repeats the experiment a second time using the same quantities of zinc and dilute sulfuric acid.

He uses a lump of zinc instead of zinc powder.

He does **not** change any other conditions.

The rate of the reaction **decreases**.

Explain why.

Use ideas about the particle model.

.....

.....

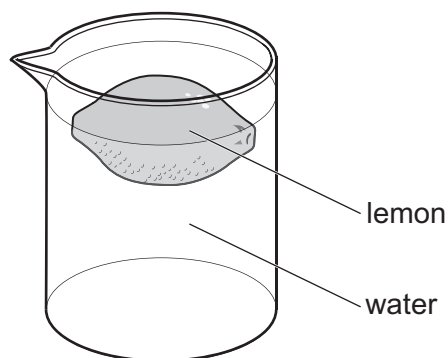
.....

.....

[2]

12 Blessy investigates floating and sinking.

She puts a whole lemon in a beaker of water.



The whole lemon floats.

Water has a density of 997 kg/m^3 .

Whole lemon has a density of 984 kg/m^3 .

Look at the table showing the density of different parts of a lemon.

part of lemon	density in kg/m^3
peel	956
segment	998
seed	925

(a) Predict which parts of the lemon float when in water.

.....

Explain your answer.

.....

.....

[1]

- (b) Blessy wants to find out if the different parts of a lemon will float or sink in salt solution.

She predicts that as the concentration of salt solution increases eventually all three parts of the lemon will float.

Write a plan for her investigation.

.....

.....

.....

.....

.....

..... [2]

13 Elements and compounds either have a giant structure or a simple structure.

The table shows information about five substances.

Complete the table to identify if the structure of each substance is giant or simple.

substance	melting point	boiling point	does it conduct electricity	structure (giant or simple)
J	low	low	no	
K	high	high	yes	
L	low	low	no	
M	high	high	no	
N	high	high	no	

[2]

14 Ahmed describes an electrical circuit.

My circuit is a complete series circuit with:

- one battery of cells
- two lamps
- one open switch
- one variable resistor
- one meter to measure the current in the whole circuit
- one meter to measure the voltage across the variable resistor.

Draw his circuit using conventional symbols.

The battery of cells has been drawn for you.



[4]

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The Periodic Table of Elements

Group																											
1	2							3	4	5	6	7	8														
1 H hydrogen 1														2 He helium 4													
3 Li lithium 7 4 Be beryllium 9 11 Na sodium 23 12 Mg magnesium 24 19 K potassium 39 20 Ca calcium 40 27 Co cobalt 59 28 Ni nickel 58 35 Br bromine 80 36 Kr krypton 84 43 Tc technetium 98 44 Ru ruthenium 101 51 Sb antimony 122 52 Te tellurium 128 59 Pr praseodymium 141 60 Nd neodymium 144 67 Ho holmium 165 68 Er erbium 167 75 Re rhenium 186 76 Os osmium 190 83 Bi bismuth 209 84 Po polonium — 91 Pa protactinium 231 92 Th thorium 232 99 Es einsteinium — 100 Fm fermium — 107 Bh bohrium — 108 Hs hassium — 115 Mc moscovium — 116 Lv livermorium — 123 N nitrogen 14 124 O oxygen 16 131 I iodine 127 132 Xe xenon 131 139 La lanthanum 139 140 Ce cerium 140 147 Tm thulium 169 148 Yb ytterbium 173 155 Lu lutetium 175 156 Er erbium 167 163 Ga gallium 70 164 Ge germanium 73 171 As arsenic 75 172 Se selenium 79 179 Sb antimony 122 180 Te tellurium 128 187 Pb lead 207 188 Bi bismuth 209 195 Tl thallium 204 196 Po polonium — 201 Fr francium — 202 Ra radium — 209 Cn copernicium — 210 Nh nihonium — 217 Uu 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Cambridge Lower Secondary Checkpoint

SCIENCE

0893/01

Paper 1

October 2023

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **18** pages.

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme. Unless specified all marking points are independent.

Annotations and abbreviations

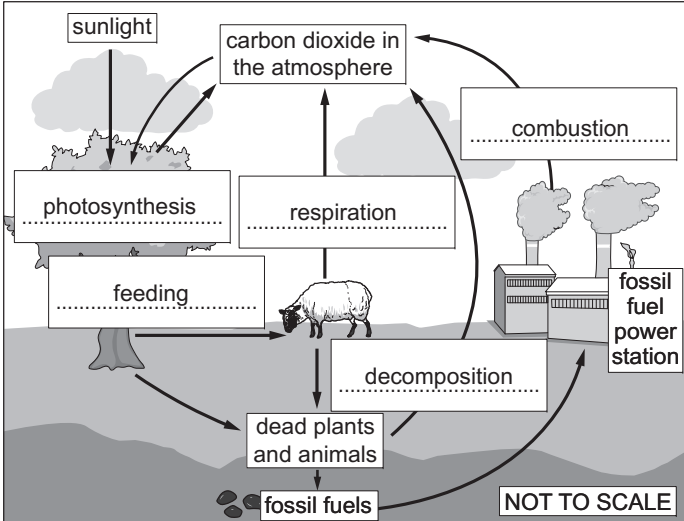
/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	(root) hair (cell)	1	
1(b)	large surface area	1	Accept long and thin or elongated Note thin alone or long alone is not sufficient Accept thin membrane or short diffusion pathway Ignore does not contain chloroplasts
1(c)	xylem (vessel)	1	Accept xylem tube or zylem
1(d)	magnesium (ion)	1	Ignore nitrates or nitrogen or phosphorus

Question	Answer	Marks	Further Information
2	any three from M1 heat (dilute) solution (of copper sulfate) M2 to make a concentrated solution or to make a saturated solution or until first appearance of crystals or solid M3 let (hot) solution cool or leave hot solution to evaporate (for a few days) M4 filter off crystals	3	each correct answer = 1 mark Note all marks can be awarded from an appropriate labelled diagram Ignore any references to how the copper sulfate solution is made. Accept for M1 evaporate solution or leave solution to evaporate Note heat or boil solution until all water evaporates gets M1 only Accept for M2 (evaporate or heat) to get half of the volume of solution Accept for M1 and M2 heat solution until first appearance of crystals = 2 marks Accept for M1 and M2 evaporate water from solution until a saturated solution is made = 2 marks Accept for M1 and M2 leave to evaporate in a warm place = 2 marks Accept for M4 and M5 pick out crystals and dry between pieces of filter paper = 2 marks

	M5 wash crystals with a small amount of cold water or dry crystals between filter paper		Accept alternative way of making crystals • hang copper sulfate crystal into copper sulfate solution = 1 mark • copper sulfate solution used is saturated or concentrated = 1 mark • leave for days (for crystal to grow) = 1 mark
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Question	Answer	Marks	Further Information
3(a)	a measure of how cold something is ☐ a measure of how hot something is ☐ a measure of the thermal energy of a substance ☒ measured in MJ ☒ transferred by conduction ☒	2	all three correct ticks = 2 marks two correct ticks = 1 mark four ticks and three correct ticks = 1 mark four ticks and two correct ticks = 0 marks five ticks = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
3(b)	any two from (idea that) the temperature is negative when water is a solid (idea that) if the temperature is 0 (°C) water is a liquid or if the temperature is 0 (°C) water is a solid (idea that) the temperature is positive when water is a liquid or the temperature is positive when water is a gas	2	each correct answer = 1 mark Do not accept temperature is negative or zero if water is a gas = max 1 mark for the question Accept ice for solid Do not accept temperature is positive if water is a solid = max 1 mark for the question Accept (idea that) the temperature is 0° C at the melting point where solid is changing into liquid or the temperature is 0° C at the freezing point where liquid is changing into ice

Question	Answer	Marks	Further Information
4(a)	 The diagram illustrates the carbon cycle. Sunlight is shown entering a tree, which is labeled 'photosynthesis'. An arrow points from the tree to a box labeled 'carbon dioxide in the atmosphere'. A sheep is shown eating from a tree, labeled 'feeding'. An arrow points from the sheep to a box labeled 'respiration', which then points to the 'carbon dioxide in the atmosphere' box. A box labeled 'decomposition' has an arrow pointing to it from 'dead plants and animals' (which are shown as a pile of leaves and a dead animal). An arrow points from 'decomposition' to 'fossil fuels' (represented by coal and oil icons). An arrow points from 'fossil fuels' to a 'fossil fuel power station' (a factory with smokestacks). An arrow points from the power station to a box labeled 'combustion', which then points to the 'carbon dioxide in the atmosphere' box. A box labeled 'NOT TO SCALE' is at the bottom right of the diagram.	3	all five correct = 3 marks three or four correct = 2 marks one or two correct = 1 mark Accept respiration for decomposition
4(b)	the breakdown of dead material (to release carbon dioxide)	1	Accept digests dead material or feeding on dead materials Accept decay or rot for breakdown Accept waste material for dead material Ignore eaten by microorganisms is not sufficient Ignore decomposition or decompose Note carbon dioxide is released is not sufficient or breakdown of material is not sufficient or breakdown of organisms is not sufficient

Question	Answer	Marks	Further Information
4(c)	any one from sea level change flooding drought extreme weather events species loss or killing of species or extinction (global) increase in (average) temperature	1	Accept named examples of extreme weather events, e.g. heatwave, cyclone or weather very hot Ignore destroying animals or plants Accept global warming or named examples of the consequence of global warming, e.g. polar ice caps melting or glaciers retreating or melting or flowering time change / changes in rainfall patterns Ignore unqualified ice melting Ignore greenhouse effect Ignore changes in temperature Ignore Earth is warm but accept Earth is warmer Do not accept acid rain or ozone depletion

Question	Answer	Marks	Further Information
5(a)	Mia (no marks) (idea of) decision based on (scientific) evidence or using evidence from a secondary source	1	Note if Aiko chosen = 0 marks Accept decision based on research or decision based on data or based on scientific or research study Accept Aiko's comment is an opinion or Mia's ideas include a fact
5(b)	smoking or drugs	1	Accept any correct idea not linked to diet, e.g. alcohol or genetics/genes (of mother or fetus) or age (of mother) or health (of mother) or multiple births etc. or medication of mother e.g. antibiotics or oxygen intake of fetus Accept emotional status of mother or stress Ignore diet or level of activity or life-style unqualified or fitness

Question	Answer	Marks	Further Information
6(a)	(mass) extinction	1	Accept named mass extinction, e.g. extinction of dinosaurs Ignore kills animals and plants Accept fires or dust clouds or air borne debris or sunlight being unable to penetrate to surface or shockwaves or seismic shaking or earthquakes or tsunami or heat radiation or wind blasts or acid rain or decrease in photosynthesis Ignore climate change and effects of climate change or destroying ecosystem or loss of habitat or global warming or explosions or temperature decrease Do not accept reference to formation of new planet or the Moon
6(b)(i)	(steel ball) asteroid (sand) (surface of the) Earth	2	each correct answer = 1 mark Accept named parts of the surface of the Earth, e.g. soil or crust or land Do not accept sea or ocean

Question	Answer	Marks	Further Information
6(b)(ii)	any two from throw the (steel) ball use a (steel) ball with more mass use a (steel) ball with a larger diameter drop the (steel) ball from a higher height increase density of ball	2	each correct answer = 1 mark Accept use more force when releasing the ball Accept heavier ball or more weight Accept bigger ball or larger ball Ignore use a higher angle Ignore more sand or increasing surface area Ignore idea of dropping more than one ball or dropping same ball several times Ignore change the sand

Question	Answer	Marks	Further Information
7(a)	cancel or cancelled or cancellation	1	Accept (complete) destructive interference Accept interference cancellation Ignore neutralisation
7(b)	A	1	more than one letter = 0 marks
7(c)	B	1	more than one letter = 0 marks

Question	Answer	Marks	Further Information							
8(a)	contains X and Y chromosomes	1	Accept contains green and yellow balls Accept contains two different chromosomes or contains two different coloured balls Accept contains XY chromosomes Ignore reference to number of chromosomes or balls							
8(b)(i)	<table><tr><td>(male or female offspring)</td></tr><tr><td>female</td></tr><tr><td>male</td></tr><tr><td>female</td></tr><tr><td>male</td></tr><tr><td>female</td></tr><tr><td>female</td></tr></table>	(male or female offspring)	female	male	female	male	female	female	1	all correct for the mark Accept boy for male and girl for female
(male or female offspring)										
female										
male										
female										
male										
female										
female										
8(b)(ii)	(idea of) not enough data (to be reliable)	1	Accept idea that the sample size is too small or did not use all the balls Accept only did it 6 times Accept need to repeat more times (for reliability)							

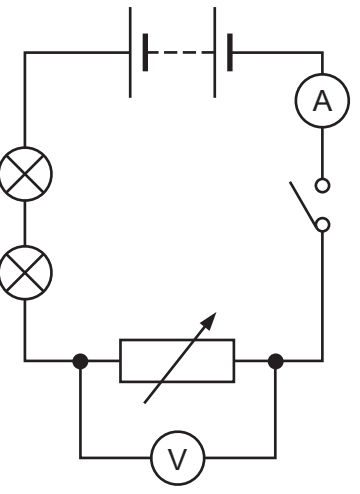
Question	Answer	Marks	Further Information
9	a sodium atom loses an electron	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

Question	Answer	Marks	Further Information
10(a)	The albatross with longer wings pass on their genes to their offspring. ☒ There is no variation in the length of albatross wings. ☐ The albatross do not compete for food. ☐ The length of the wings decreased over many generations. ☐ The albatross with longer wings are more likely to survive. ☒	2	each correct tick = 1 mark three boxes ticked and two correct = 1 mark three or more boxes ticked and one correct = 0 marks four or more boxes ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
10(b)	decreased environmental extinct	3	each correct answer = 1 mark Accept climate or habitat (destruction) or food supply or hunting or pollution or death rate is more than birth rate or increased number of predators Ignore weather Accept endangered

Question	Answer	Marks	Further Information
11(a)	$\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$	2	reactants in either order = 1 mark products in either order = 1 mark Ignore any names of chemicals
11(b)(i)	line starting at origin with an initial steeper gradient same volume of gas produced	2	each correct answer = 1 mark Accept a tolerance of $\pm$ one small square for (0,0) Accept two straight lines or dot-to-dot type graphs Note the gradient must never increase over time Accept a tolerance of $\pm$ half a small square Do not accept any line which goes above 41 cm ³ for this marking point
11(b)(ii)	(lump has) smaller surface area fewer collisions	2	each correct answer = 1 mark Accept ora if specified Accept fewer exposed particles Accept collisions less often or less chance of collisions or less collision frequency = 2 marks

Question	Answer	Marks	Further Information
12(a)	(float) peel and seed and (explanation) density less than water	1	both answers and explanation correct for the mark
12(b)	make different concentrations of salt solution or add different amounts of salt to water drop each part of the lemon into the salt solution and see if it floats	2	each correct answer = 1 mark Note mark independently Accept each part of the lemon into the salt solution and see the results Ignore drop (whole) lemon into salt solution and see if it floats

Question	Answer		Marks	Further Information
13	substance	structure (giant or simple)	2	all five correct = 2 marks two, three or four correct = 1 mark one correct = 0 marks
	J	simple		
	K	giant		
	L	simple		
	M	giant		
	N	giant		

Question	Answer	Marks	Further Information
14		4	M1 two lamps in series and one open switch = 1 mark M2 ammeter in series and must measure the total current = 1 mark M3 variable resistor and no significant gaps in circuit (by eye) = 1 mark Accept for M3 two minor gaps in circuit M4 voltmeter across any attempt at a variable resistor = 1 mark Note correct symbols must be drawn for M1, M2 and M3 but ignore line going through ammeter, lamp and voltmeter symbol Accept incorrect variable resistor for M4 but symbol for voltmeter must be correct Ignore missing junction symbols

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Cambridge Lower Secondary Checkpoint

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SCIENCE

0893/02

Paper 2

October 2023

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

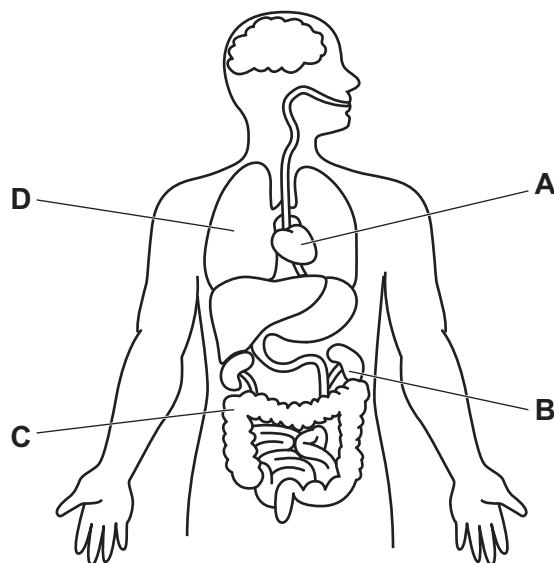
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

- 1 The diagram shows the position of some organs in the human body.



- (a) Circle the letter of the organ that is part of the human excretory (renal) system.

A

B

C

D

[1]

- (b) Complete these sentences about the human excretory (renal) system.

The function of the human excretory (renal) system is to the blood to remove urea.

The urea is then excreted in a liquid called

[2]

- (c) Organs are made of cells.

Cells contain chromosomes.

Name the chemical from which chromosomes are made.

..... [1]

- 2 Look at the elements in Group 1 from the Periodic Table.

The elements are in the same order as the Periodic Table.

element
lithium
sodium
potassium
rubidium
caesium
francium

- (a) Describe how the melting points of the Group 1 elements change down the group.

..... [1]

- (b) Name the **most** reactive element in Group 1.

..... [1]

- (c) An element in Group 1 reacts with dilute hydrochloric acid.

Circle the gas made in this reaction.

carbon dioxide

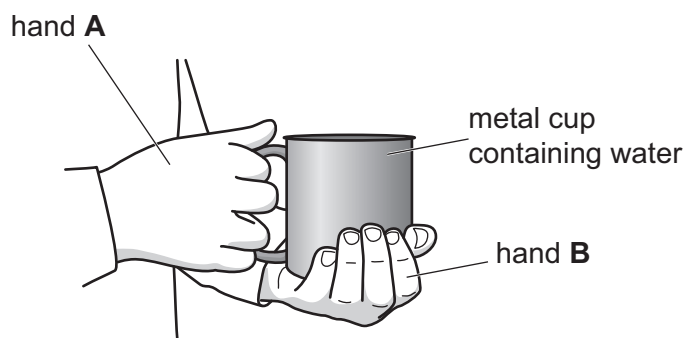
chlorine

hydrogen

oxygen

[1]

- 3 Mike holds a metal cup containing water.



- (a) The water in the metal cup is at a higher temperature than both of his hands.

Describe what happens to the thermal energy in the water.

..... [1]

- (b) Mike pours the water out of the metal cup.

He adds ice and water to the metal cup.

Describe what Mike feels with hand B compared to hand A.

..... [1]

- 4 The atoms in a molecule of water are joined together by covalent bonds.

- (a) What is a covalent bond?

..... [2]

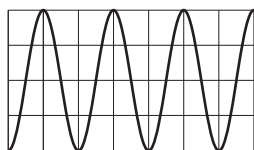
- (b) The formula for a molecule of ethane is C_2H_6 .

How many atoms are bonded together in one molecule of ethane?

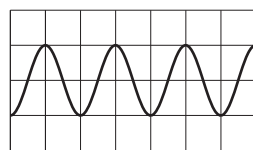
..... [1]

5 Priya compares different sound waveforms.

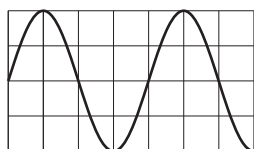
All the waveforms are drawn to the same scale.



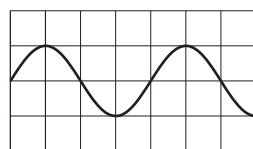
A



B



C



D

(a) Which **two** waveforms have the **lowest** amplitude?

..... and

[1]

(b) Which **two** waveforms have the **lowest** frequency?

..... and

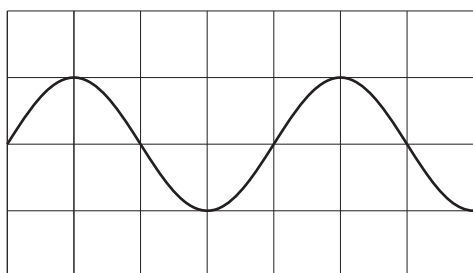
[1]

(c) Which **two** waveforms have the **highest** pitch?

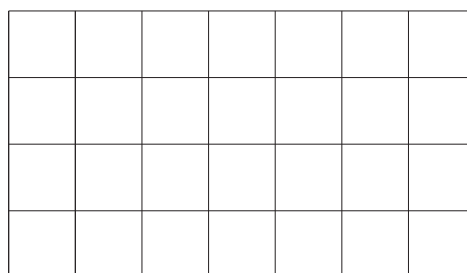
..... and

[1]

(d) Draw a waveform on the grid with a **greater** loudness than waveform E.

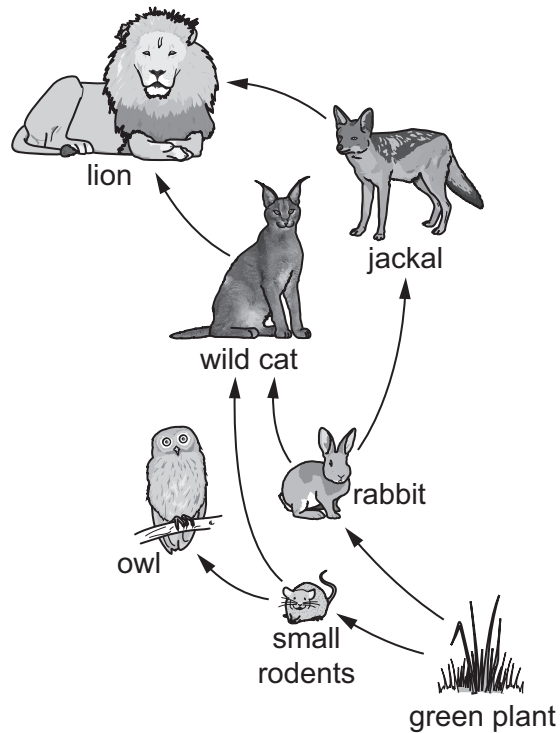


E



[1]

6 Look at the food web for a habitat.



NOT TO SCALE

(a) A disease decreases the number of small rodents in the habitat.

The number of owls decreases but the number of wild cats stays the same.

Complete the sentences to explain why.

The number of owls decreases because

.....

The number of wild cats stays the same because

.....

[2]

(b) Explain why green plants need the Sun to survive.

.....

.....

.....

[3]

- 7 An object has volume, mass and density.

The object has a volume of 28 cm^3 .

The mass of the object is 222 g.

Calculate the density of the object.

..... g/cm^3 [2]

- 8 The diagram shows part of the Periodic Table.

1 H							2 He
3 Li	4 Be	5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca						

- (a) Which element in the table has the lowest number of protons in its atom?

.....

[1]

- (b) Identify **two** elements from the table that are in the same period as the element Mg.

..... and

[1]

- (c) Name **one** element from the table that has the same chemical properties as the element Ar.

.....

[1]

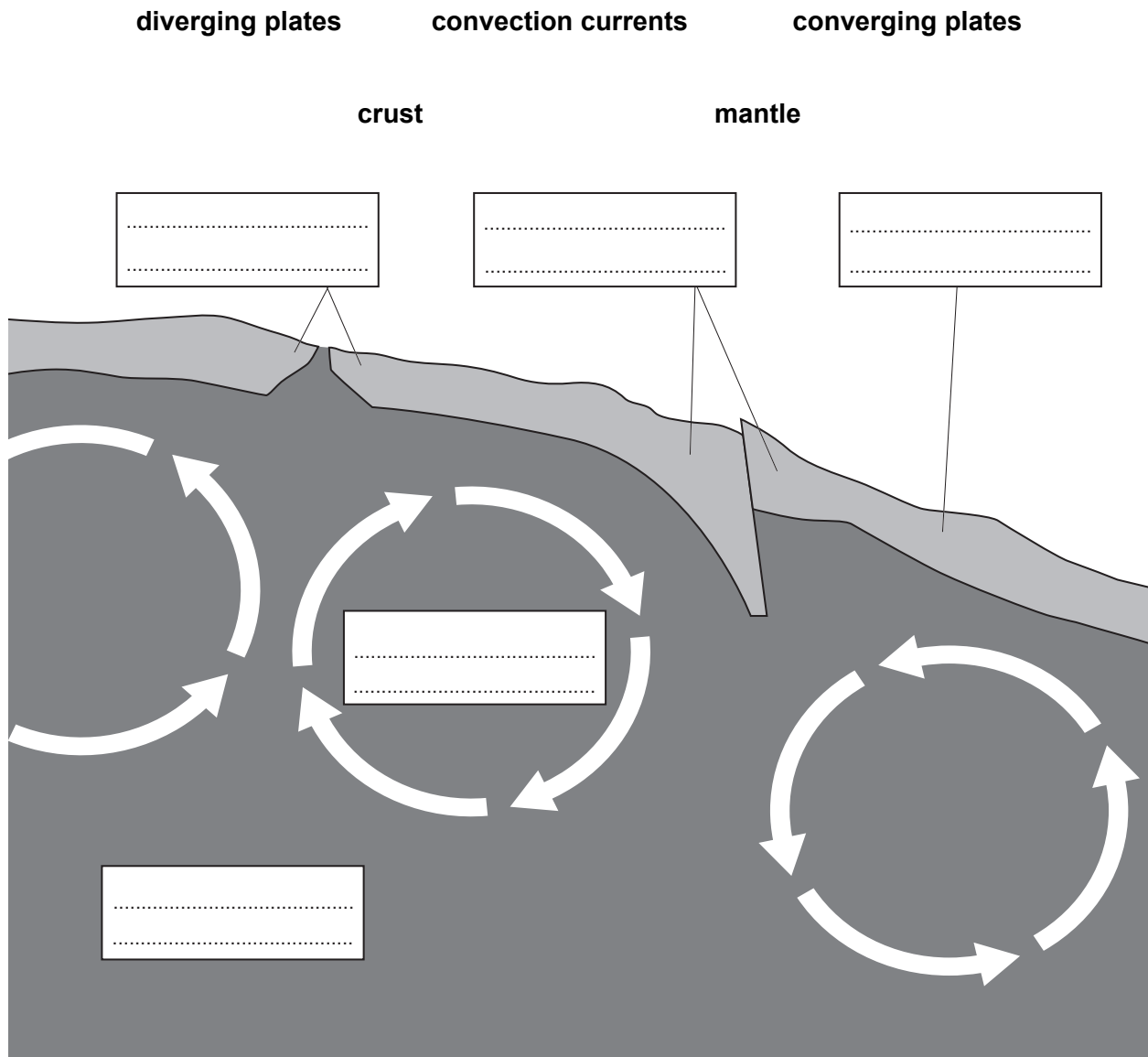
9 Mia is learning about tectonic plates.

She knows that:

- diverging tectonic plates move away from each other
- converging tectonic plates move towards each other.

(a) Label the diagram.

Choose words from the list.



[2]

(b) Mia joins a map of South America to Africa.



Explain why the appearance of the continental coasts is evidence for tectonic plates.

.....

.....

..... [2]

(c) Write down **one other** piece of evidence for tectonic plates.

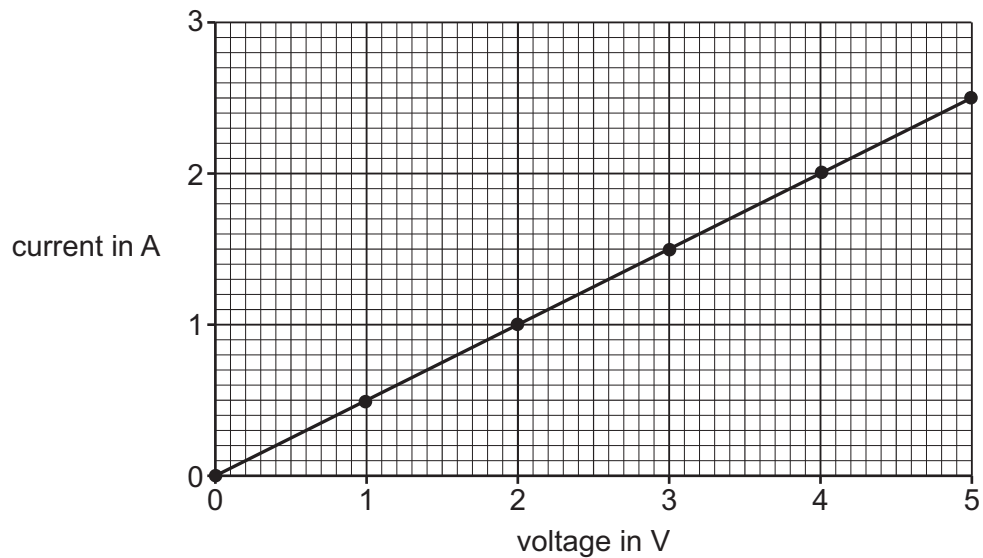
.....

..... [1]

10 Oliver connects a resistor in an electrical circuit.

He measures the current as he increases the voltage across the resistor.

Oliver draws a graph.



(a) Current is measured in A (amps) and voltage is measured in V (volts).

Write down the unit of resistance.

..... [1]

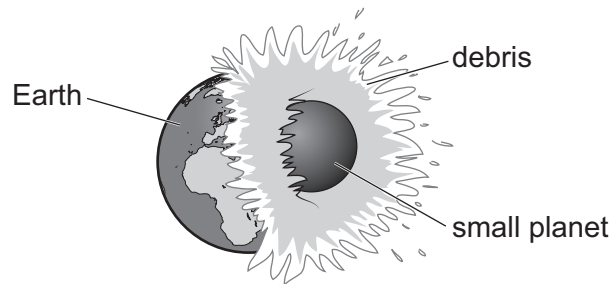
(b) Calculate the resistance of the resistor.

Include the equation used to calculate resistance in your answer.

resistance = [2]

- 11 Scientists believe that the Moon was formed after a collision between the Earth and another small planet.

This is called the collision theory for the formation of the Moon.



The debris from the collision collected to form the Moon.

- (a) Chen collects information about the elements found on the Earth and on the Moon.

element	percentage of element found on the	
	Earth	Moon
oxygen	45.3	44.7
silicon	22.0	22.5
magnesium	2.6	2.3
iron	6.0	8.3
calcium	3.6	3.1

- (i) Most of the information supports the collision theory.

Explain how most of this information supports the collision theory.

.....
 [1]

- (ii) There is a comparison of one element that does **not** support the collision theory.

Write down the name of this element.

..... [1]

- (b) Suggest **two other** pieces of evidence Chen collects to support the collision theory.

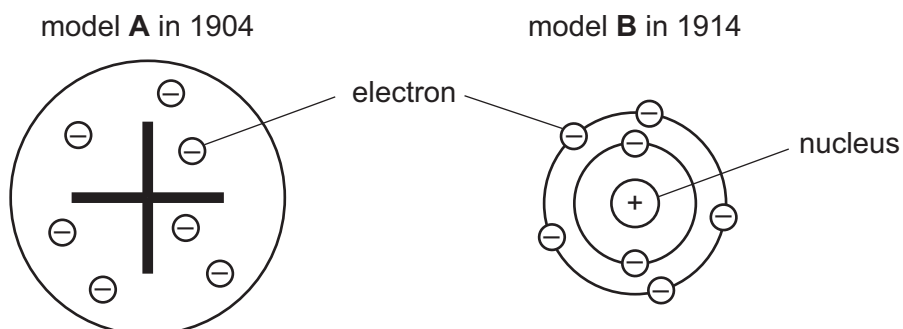
1

2

[2]

12 Theories about the structure of the atom have developed over time.

Look at the models of an atom of nitrogen.



(a) Describe **one similarity** between model **A** and model **B**.

.....
 [1]

(b) Describe **one difference** between model **A** and model **B**.

.....
 [1]

(c) Model **B** is still used today.

Suggest **one strength** and **one limitation** of using model **B**.

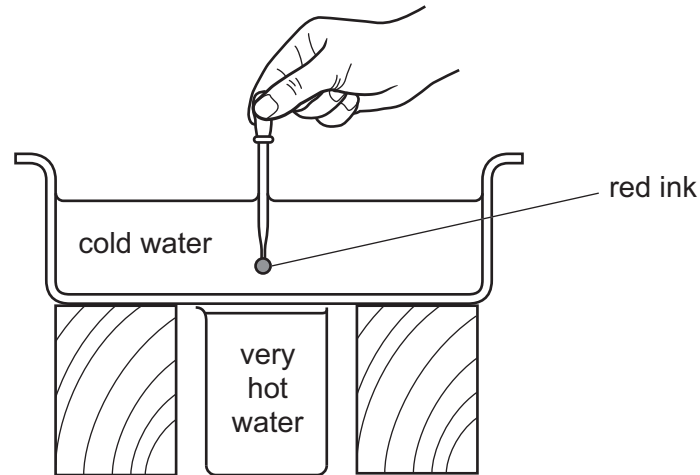
strength

 limitation

 [2]

13 Yuri investigates convection.

He adds a drop of red ink to the cold water as shown in the diagram.



(a) Complete the sentence to suggest a testable hypothesis for **this** investigation.

I predict that the red ink will move

because

[1]

(b) Complete the table about **safety risks** and the **control of risks** in this investigation.

safety risk	control of risk
very hot water may burn skin	
red ink may irritate skin	
.....	use plastic beaker instead of glass beaker

[3]

14 Lily investigates variation in tomatoes.

Lily:

- measures the mass of different tomatoes to the nearest whole gram
- classifies the tomatoes into different groups based on their masses.

Lily writes about her results.

There were 3 tomatoes with a mass more than 105 g.

There were 2 tomatoes with a mass between 76 g and 80 g.

There were 6 tomatoes with a mass between 101 g and 105 g.

There were 8 tomatoes with a mass between 86 g and 90 g.

There were 9 tomatoes with a mass between 91 g and 95 g.

There were 12 tomatoes with a mass between 96 g and 100 g.

There were 4 tomatoes with a mass between 81 g and 85 g.

(a) (i) Complete the table of results by writing the:

- unit for the mass range
- number of tomatoes in each mass range.

mass range in	number of tomatoes in mass range
76 – 80	
81 – 85	
86 – 90	
91 – 95	
96 – 100	
101 – 105	
more than 105	

[2]

(ii) What is the best way to present the data in the table?

..... [1]

(b) Gardeners add nitrates to the soil to help tomato plants grow.

The nitrates are used by the plants to make a substance needed for growth.

Name this type of substance.

..... [1]

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The Periodic Table of Elements

Group																		
1	2							7	8									
				1 H hydrogen 1						2 He helium 4								
		Key atomic number atomic symbol name relative atomic mass																
3 Li lithium 7	4 Be beryllium 9									5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20			
11 Na sodium 23	12 Mg magnesium 24									13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40			
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —
lanthanoids																		
57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175				
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —				
actinoids																		



Cambridge Lower Secondary Checkpoint

SCIENCE

0893/02

Paper 2

October 2023

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **16** pages.

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme.
Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	B	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
1(b)	filter urine	2	each correct answer = 1 mark Accept clean or purifies
1(c)	DNA	1	Accept deoxyribonucleic acid Ignore genes or nucleic acid

Question	Answer	Marks	Further Information
2(a)	(melting point) decreases	1	
2(b)	francium	1	Accept Fr Accept francium circled in the table, but answer line takes precedence
2(c)	hydrogen	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

Question	Answer	Marks	Further Information
3(a)	any one from (idea that) thermal energy transfers (from water) to cup (idea that) thermal energy transfers (from water) to his hand(s) (idea that) thermal energy transfers (from water) to air	1	Accept heat or 'it' for thermal energy throughout Accept (idea that) thermal energy transfers from cup to hands or thermal energy heats the hand Accept (idea that) thermal energy transfers from cup to air Accept (idea that) thermal energy transfers from hand to air Accept (idea that) thermal energy transfers from hotter region to colder region or thermal energy dissipates Accept thermal energy of water decreases
3(b)	hand B is colder	1	Note answer must be comparative Note assume answer refers to hand B unless hand A is specified

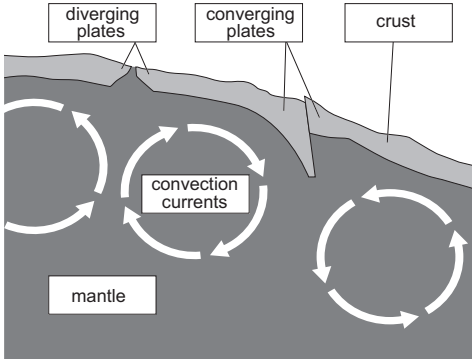
Question	Answer	Marks	Further Information
4(a)	(idea of) sharing electrons (between atoms) (idea of) a pair of electrons (being shared)	2	each correct answer = 1 mark Accept two electrons shared (between atoms) = 2 marks Accept a shared pair of electrons (between atoms) = 2 marks Ignore comments about metals and non-metals, intermolecular forces or ions
4(b)	8	1	

Question	Answer	Marks	Further Information
5(a)	B (and) D	1	both correct either order = 1 mark
5(b)	C (and) D	1	both correct either order = 1 mark
5(c)	A (and) B	1	both correct either order = 1 mark
5(d)	waveform drawn with greater amplitude	1	Accept waveform drawn on the same grid as E, but answer grid takes precedence Ignore any change in frequency Note a smooth curve is not necessary as long as all the amplitudes are higher than E

Question	Answer	Marks	Further Information
6(a)	(The number of owls decrease because) they (only) feed on small rodents (The number of wild cats stays the same because) wild cats will eat (more) rabbits	2	each correct answer = 1 mark Accept they have no food or have no alternative food or have less food Ignore the small rodents died or small rodents decrease in number Accept they have another food source or can eat rabbits and rodents Ignore the rabbit does not decrease or rabbits are not affected or number of rabbits stays the same
6(b)	source of light for photosynthesis (use light energy) to make carbohydrates	3	each correct answer = 1 mark Accept source of energy or (source of) sunlight or (source of) light Ignore provides heat Accept a description of photosynthesis Accept (use light energy) to make glucose or make starch Accept just make own food

Question	Answer	Marks	Further Information
7	(density =) mass ÷ volume 7.9(2857)	2	Accept 222÷28 Accept correct rounding, e.g. 7.93 or 7.929 or 7.9286 Accept correct answer with no working = 2 marks Accept 8 or 7.92 = 1 mark

Question	Answer	Marks	Further Information
8(a)	H	1	Accept hydrogen
8(b)	any two from Na, Al, Si, P, S, Cl, Ar	1	Accept correct names Note if an incorrect element is given = 0 marks
8(c)	neon or helium	1	Accept Ne or He

Question	Answer	Marks	Further Information
9(a)	 The diagram illustrates the process of plate tectonics. It shows a cross-section of the Earth's upper mantle and crust. The mantle is depicted with three circular convection currents, indicated by white arrows. The top layer is the crust, shown in a lighter shade. Above the crust, tectonic plates are shown. One pair of plates is labeled 'diverging plates' and is moving apart, while another pair is labeled 'converging plates' and is moving together. The entire system is labeled 'crust' and 'mantle'.	2	all five labels correct = 2 marks two, three or four labels correct = 1 mark one label correct = 0 marks Accept crust instead of the label for diverging or converging plates
9(b)	(idea that) the shapes (of the coastlines) of South America and Africa are complementary or (idea that) the coastlines fit together like (pieces of) a jigsaw (idea that) this suggests that they were once joined together and have moved apart as the tectonic plates move apart	2	each correct answer = 1 mark Accept (idea that) the coastlines fit together (almost perfectly) Accept (idea that) the continents fit together (almost perfectly) Note coastlines match is not sufficient Do not accept they are the same shape Note have moved apart is not sufficient but Accept have moved apart as tectonic plate move Accept continental drift as tectonic plates move or (continents) split apart as tectonic plates move Note once joined together or once part of Pangea is not sufficient Ignore move apart and break

Question	Answer	Marks	Further Information
9(c)	any one from (location of) volcanoes (location of) earthquakes fossil records alignment of magnetic material in the crust	1	 Ignore same fossil fuels Accept same rock types Ignore mountain ranges and mountain building

Question	Answer	Marks	Further Information
10(a)	ohm or Ω	1	
10(b)	(resistance =) $\frac{\text{voltage}}{\text{current}}$ 2	2	each correct answer = 1 mark Accept $\frac{V}{I}$ or $V = IR$ Accept symbols or words or units or pairs of numbers from the graph in any correct rearrangement, e.g. volts $\div$ amps or 4 and 2 seen in calculation Note correct answer only = 1 mark Ignore units

Question	Answer	Marks	Further Information
11(a)(i)	(idea that) the percentage of an element found on the Earth is similar to the percentage of an element found on the Moon	1	Accept Earth and Moon have similar compositions (of elements) Accept the elements are similar on Earth and Moon or most elements found on both Earth and Moon or Moon has most of the elements found on Earth Ignore percentage of oxygen on Earth and Moon is high
11(a)(ii)	iron	1	
11(b)	any two from find the percentage of more elements (on the Earth and on the Moon) presence of water on Moon both Moon and Earth spin Moon is less dense than the Earth	2	each correct answer = 1 mark Allow find the percentage of elements (on Earth and the Moon) not named in the table Ignore presence of carbon dioxide on Moon Accept both orbit the Sun or both reflect light Accept both have cores with iron or Moon has a small iron core or both have molten cores Accept similar age of rocks on Moon and the Earth Accept similar types of rock on Earth and the Moon

Question	Answer	Marks	Further Information
12(a)	both have (7) electrons	1	Accept both have a positive (and a negative part) Accept both have a negative part Ignore references to nucleus and protons e.g. both have nucleus or both have protons
12(b)	A does not have nucleus or B has a nucleus or A does not have electrons in orbits or shells or B has electrons in orbits or shells	1	Note assume unqualified answers refer to model A Ignore size of nucleus Accept A has no shells or orbits or B has shells or orbits Accept circles or rings for shells Note electrons in different positions is not sufficient

Question	Answer	Marks	Further Information
12(c)	(strength) any one from shows the relative positions of the nucleus and/or electrons shows the positive and negative part of the atom shows particles that are too small to be seen shows the space in an atom shows electrons in shells or shows electronic structure or shows number of electrons (limitation) any one from does not show the movement of the electrons the model is two dimensional or model is not in three dimensions does not show particles within the nucleus or does not show protons or does not show neutrons the model is not to scale	2	each correct answer = 1 mark Accept gives a clear understanding of the atom or provides an image and understanding of an atom Accept show the charge of particles Accept shows (outermost) shell or orbits or shows electrons or shows how many electrons in a shell Accept can tell which group it is in or can tell which period it is in Accept do not know the exact size of the atom or not same size as real atoms Accept not proven as it is a theory

Question	Answer	Marks	Further Information
13(a)	up or upwards and cold water has been warmed or the warmed water rises or (the warmed water has an upward) convection current	1	Note the mark should be awarded wherever it is written both correct for the mark Accept up then down Ignore away from the hot water or up and down Ignore heat rises
13(b)	(very hot water) use heat proof gloves (red ink) wear gloves or do not use poisonous red ink (use plastic ...) glass may break or may cut hand on glass	3	each correct answer = 1 mark Accept safety gloves or protective gloves or thick gloves or use tongs Ignore reduce water temperature Accept use a food colour Note use another ink colour unqualified is not sufficient Ignore use a pipette or use a syringe Accept glass may crack Ignore the glass might explode or beaker may be hot

Question	Answer	Marks	Further Information																
14(a)(i)	<table><tr><td>mass range in g or grams</td><td>number of tomatoes in mass range</td></tr><tr><td>(76 – 80)</td><td>2</td></tr><tr><td>(81 – 85)</td><td>4</td></tr><tr><td>(86 – 90)</td><td>8</td></tr><tr><td>(91 – 95)</td><td>9</td></tr><tr><td>(96 – 100)</td><td>12</td></tr><tr><td>(101 – 105)</td><td>6</td></tr><tr><td>(more than 105)</td><td>3</td></tr></table>	mass range in g or grams	number of tomatoes in mass range	(76 – 80)	2	(81 – 85)	4	(86 – 90)	8	(91 – 95)	9	(96 – 100)	12	(101 – 105)	6	(more than 105)	3	2	g or grams in heading = 1 mark all numbers correct = 1 mark
mass range in g or grams	number of tomatoes in mass range																		
(76 – 80)	2																		
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(101 – 105)	6																		
(more than 105)	3																		
14(a)(ii)	bar chart	1	Accept histogram or frequency polygon or bar graph Ignore graph or frequency chart Do not accept line graph, scatter plot or pie chart																
14(b)	protein	1	Ignore chlorophyll																